1.6 Nomenclature

Giza, Inimene, Emberi, Manna,
Duine, Umano …….

In all probability these words must be new to you…but they all mean "Human" in different foreign languages! There are presently more than 6000 languages in the world and an animal can be named in more than 6000 ways! Unfortunately it is impossible for anyone to have a good functioning knowledge of most languages and hence there arises a need for a universally accepted scientific naming system for all organisms. The process of assigning scientific names to animals or taxonomic group is called nomenclature. For example, worldwide, the scientific name *Homo sapiens* denotes *human*. Classification and grouping were done to facilitate a deeper understanding of the unique characteristics of each organism and its interrelationship among closely related species. It plays a vital role in the arrangement of known species based on their similarities and dissimilarities. Numerous characters such as morphology, genetic information, habitat, feeding pattern, adaptations, evolution, etc., are examined before an organism is named.

One of the primary responsibilities of systematic biology is the development of biological nomenclature and classification. Nomenclature is not an end to systematics and taxonomy but it is necessary in organizing information about biodiversity. Nomenclature, functions to provide names for all taxa at all levels in the hierarchy of life. Naming of the organisms is done based on the guidelines of the International Code of Zoological Nomenclature (ICZN). The scientific name ensures that each organism has only one name.

Binomial nomenclature
(L. *Bi*-two; *Nomen*-Name)

Biologists follow universally accepted principles to provide scientific names to known organisms. Each name has two components, a generic name and a specific epithet. This system of naming the organism is called **Binomial Nomenclature** which was popularised by Carolus Linnaeus and practised by biologists all over the world. Example, the National Bird (Indian Peafowl) – *Pavo cristatus,* the National Animal tiger as *Panthera* tigris, and the Tamil Nadu State bird is the common Emerald dove *Chalcophaps indica*.



Biological nomenclature derives from the binomial (or binominal) nomenclature that was originally codified in the works of Linnaeus, *Species Plantarum* (1753) and *Systema Naturae*, 10th Edition (1758). These publications are the starting points for the modern biological nomenclature in most groups of plants and animals.

If you find an animal with four legs, with two eyes, paired ear pinna, covered with fur, possessing mammary gland , which class will you position it? How will you give a binomial name, if you are the first person to discover and report that animal.

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